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MIL-STD-2189(SH)
SECTION 314-4
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MILITARY STANDARD

DESIGN METHODS FOR NAVAL SHIPBOARD SYSTEMS

SECTION 314-4

400 HERTZ POWER SYSTEM TEST PROCEDURE



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DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

Washington, DC 20362-5101

Design Methods for Naval Shipboard Systems, 400 Hertz Power System Test Procedure

1. This Military Standard is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. GENERAL AND SCOPE

1.1 General. The procedures established by MIL-STD-2189 are applicable. This section and the basic standard are to be considered as an integral single document.

1.2 Scope. This standard provides a guide for conducting 400 hertz (Hz) power system tests to determine that the 400 Hz power system is compatible with the equipment served and that the system characteristics are within the limits specified in DOD-STD-1399, section 300.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Standards. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

DOD-STD-1399, Section 300	- Interface Standard for Shipboard Systems, Section 300 Electric Power, Alternating Current (Metric).
MIL-STD-2189, Section 310-1	- Design Methods for Naval Shipboard Systems, Electrical System Load and Power Analysis for Surface Ships.
MIL-STD-2189	- Design Methods for Naval Shipboard Systems

(Copies of standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

3. DEFINITIONS

3.1 Ship operating conditions. For definitions of ship operating conditions, see MIL-STD-2189, section 310-1.

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4. GENERAL REQUIREMENTS

4.1 Test instruments. Instruments required for these tests shall have accuracies in accordance with table I. Instruments used for test measurements shall be calibrated. Instrument data and calibration dates shall be recorded. Figure 1 is a sample of the information required.

TABLE I. Instrument types and accuracies.

Type measurement	Type measuring instrument	Speed and accuracy	Parameters measured	Remarks
Short-time transient	Fast storage scope or transient peak recorder	0.5 to 800 micro-seconds ± 5 percent reading ± 1 percent 1 full scale	Peak voltage	Continuous during tests
Long-time transient (load switching)	Scope and camera plus transducer, V converter or oscilloscope or oscillograph	V = ± 5 percent of transient I = ± 5 percent F = ± 1 percent nominal	Transients: Volts (V) Current (I) Frequency (F) V and F recovery time	Taken at time load is turned on and load is turned off
Steady-state	True rms digital voltmeter (V) 3-ph wattmeter (W)* Spectrum analyzer (H) Current transducer (I) Digital frequency meter (f) Single trace or dual trace storage oscillo-scope with camera (A) *May use meter on switchboard if available.	V = ± 0.2 percent W = ± 5 percent II = ± 10 percent I = ± 5 percent F = ± 0.1 percent A = ± 0.2 percent	3-ph average volts (V) 3-ph watts (W) Frequency (F) Current (I) Voltage unbalance (V) Harmonic dis-tortion (II) Amplitude modulation (A)	Strip type on sample printer type instru-ments may be used for these readings as applicable.

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4.2 General description of tests. The tests shall be conducted with normal (battle or functional) loads on the equipment providing power to the 400 Hz power conversion equipment (PCE) primary power sources (normally the 60 Hz ship service distribution system). Tests on the 400 Hz system shall be conducted at no load, normal battle or functional load, and alternate load, or a combination of normal plus alternate loads, as applicable. During tests under the various loaded conditions, all available equipment shall be in operation at the same time to ensure equipment compatibility. Only the tests which require information to be recorded (see examples of figures 2 and 3) need to be conducted when the PCE is dedicated to a single load. The tests shall determine.

(a) Steady state voltage.
(b) Steady state frequency.
(c) Steady state line current.
(d) Steady state voltage amplitude modulation.
(e) Steady state voltage unbalance.
(f) Average phase voltage for 3 phases.
(g) Transient voltage, current, and frequency.
(h) Voltage and frequency recovery time.
(i) Maximum single harmonic.
(j) Total harmonic distortion.

During tests, equipment using 400 Hz shall be observed and any malfunction shall be noted. If no malfunction is observed, this fact shall be recorded on the information form. If characteristics such as voltage unbalance, amplitude modulation, or harmonics, exceed the limits specified in DOD-STD-1399, section 300, individual loads shall be disconnected and measurements repeated for the particular characteristic to determine if the load causing the departure from the specified limit can be isolated. Loads causing characteristic limits to be exceeded shall be identified and recorded.

4.3 Meter checks.

4.3.1 Frequency meter checks. Prior to start of tests, it shall be ascertained that the 60 Hz power source is operating at nominal 60 Hz setting. The 3-phase voltages and frequency of the input power to the power conversion equipment (motor-generator sets, inverters, and converters) at the nearest switchboard load center or panel (distribution point) supplying the equipment shall be observed and recorded. With the 400 Hz PCE operating at no load, and after the equipment has reached normal operating temperature, the frequency of the PCE shall be set at 400 Hz using a frequency standard. A check shall be made for error between the switchboard-mounted meter and the frequency standard. If the error is outside of the calibration limits of the switchboard mounted meter, the meter shall be recalibrated to bring it within the calibration limits. If the error in calibration of the switchboard mounted frequency meter cannot be corrected, the error shall be noted (see sample shown on figure 2).

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4.3.2 Voltmeter checks. Prior to start of tests, it shall be ascertained that the 60 Hz power source is operating at nominal 450 volts. With each 400 Hz PCE operating at no load, the output voltage of the equipment shall be set at 450 volts using a true root mean square voltage standard. If error is outside of the calibration limits of the switchboard-mounted meter, the meter shall be recalibrated in order to bring it within the calibration limits. If calibration of switchboard mounted voltmeter cannot be obtained, the error shall be noted (see sample shown on figure 2.)

4.4 Power conversion equiqment tests. Power conversion equipment tests shall be performed as specified in 4.4.1 through 4.4.5.

4.4.1 No load or minimal load test. Prior to starting, record the 3-phase voltages and the frequency of the input power to the PCE at the distribution point that normally feeds the equipment. Record the switchboard load center of panel number. With no load on the equipment (or minimal load on equipment provided with no load disconnect), start the unit. After the equipment has reached normal operating temperature, record the phase voltages and frequency of the PCE output. These readings shall be taken at the nearest distribution equipment fed by the equipment. For power conversion equipment provided with output voltage compensation, the readings shall be taken at the distribution equipment or bus where the line voltage has been compensated. Record the distribution equipment number. Record the readings of 400 Hz switchboard-mounted voltage and frequency meters. Indicate any error. Record information (see sample shown on figure 2.)

4.4.2 Steady-state tests. With the PCE supplying its normal load, record the 3-input phase voltages, and the input frequency record and the number of the switchboard where the readings were taken. Record the PCE output frequency and phase voltages, and the number of the switchboard where the readings were taken. For the output of the PCE, record the average line-to-line 3-phase voltage, the maximum voltage unbalance between phases, the percent voltage unbalance , and percent amplitude modulation of each phase voltage. Record the output line current in one phase, and the total kilowatt (kW) rating of the load. Record the information (see sample shown on figure 3). If voltage unbalance limits are exceeded, record all output line currents.

4.4.3 Transient tests. With the PCE supplying its normal load, suddenly remove the largest single load which can be deenergized and rapidly reenergize without damage. Allow the equipment to stabilize, then reapply the load in one step. Record the required information (see sample shown on figures 4 and 5).

4.4.4 Repeat tests. Repeat the steady-state test and the transient test with the PCE fed from the same 60-Hz source, but supplying alternate loads or a combination of normal and alternate loads, as applicable. Record the information (see samples shown on figures 3, 4, and 5).

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4.4.5 Harmonic analysis. Take the harmonic voltage measurements from the first through the fiftieth harmonic on one output phase of the PCE at no load, normal load, and alternate load, or a combination of normal and alternate loads, as applicable. Record the information (see sample on figure 6). Measure the harmonic voltages from one through thirteen on all three output phases of PCE with the equipment feeding no load, normal load, and alternate load, or a combination of normal and alternate loads, as applicable. Record the information (see sample shown on figure 7).

4.5 Line voltage regulator tests. Line voltage regulator tests shall be performed as specified in 4.5.1 through 4.5.4.

4.5.1 Power conversion equipment - line voltage regulator (LVR) no load stability tests. For each PCE that feeds LVR's, perform the following tests: with no load on the PCE other than the unloaded LVR the PCE supplies, vary the PCE voltage slowly over its maximum adjustment range while monitoring the PCE output voltage modulation. Record the maximum value of the modulation and the voltage at the time of the maximum modulation. Ensure that all LVR's supplied by the PCE are energized during the test. In the case of LVR's installed on aircraft servicing systems having available load monitors, it will be necessary to place a nominal load of 1 to 2 amperes on the regulators to close the input power switch to the regulator. Record the information (see sample shown on figure 8).

4.5.2 Steady state tests. With the LVR operating at normal load, adjust the regulator for proper output voltage in accordance with the LVR technical manual. Record the input power source, the input voltage, frequency, and line current. Record the regulator output voltage, frequency, line current, and percent voltage modulation. Repeat the procedure, except with the regulator supplied from its alternate source. Record the information (see sample shown on figure 9). Under remarks, indicate if the line voltage regulators are connected in 3-phase, and if connected wye or delta.

4.5.3 Transient tests. With the regulator supplying its normal load, suddenly remove the largest single load which can be deenergized and rapidly reenergize without damage. Record the voltage transient and recovery time and the frequency transient and recovery time. Identify the load removed and record the full load line current of the load. After allowing the regulator to stabilize, reapply the load and record the voltage and frequency transient and recovery time. Record the information (see sample shown on figure 10).

4.5.4 LVR harmonic analysis tests. Take harmonic measurements on the output of 400 Hz voltage regulators. Record harmonic voltages one through thirteen. Measurements shall be made at no load and at normal load. Record the information (see sample shown on figure 11).

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5. DETAILED REQUIREMENTS

Not applicable.

6. NOTES

6.1 Intended use. This standard is used to assure that the ships 400 Hz systems characteristics are within established limits and are compatible with equipment served.

6.2 Subject term (key word) listing.

Harmonic analysis
Line voltage regulator tests
Meter checks
Test instruments
Voltmeter checks

Preparing activity:
Navy - SH
(Project 1990-N056)

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INSTRUMENT DATA

SHIP NAME AND NO. _____

Instrument name and manufacturer	Serial no.	Range	Accuracy	Date of last calibration	Swbd No.	Remarks

FIGURE 1. Sample information for instrument on data.

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Output Voltage after start	AB						
	BC						
	CD						
Output Frequency							
Installed meters		Volts	Percent error	Volts	Percent error	Volts	Percent error
Voltmeter	AB						
	BC						
	CA						
Frequency Meter		Hz		Hz		Hz	

Remarks:

FIGURE 2. Sample inforamtion for power conversion equipment - no load test.

POWER CONVERSION EQUIPMENT - NO LOAD TEST

SHIP NAME AND NO._____

PCE No:							
kW Rating							
Type Power (Output)							
60-Hz Swbd. No. (Normal Input)							
Input voltage prior to start	AB						
	BC						
	CA						
Input Frequency							
		Normal	Alternate	Normal	Alternate	Normal	Alternate
400-Hz Swbd. No.							

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POWER CONVERSION EQUIPMENT - STEADY-STATE TEST

SHIP NAME AND NO. _____

		PCE NO. Normal	Alternate	PCE NO. Normal	Alternate	PCE NO. Normal	Alternate
Swbd or Panel No.			XXXXXXXXX		XXXXXXXXX		XXXXXXXXX
Frequency	Hertz						
	Phase AB						
	Phase BC						
	Phase CA						
Swbd or Panel No.							
Frequency	Hertz						
	Phase AB						
Output Volts	Phase BC						
	Phase CA						
Average Line to line 3-ph Volts							
Maximum Voltage Unbalance	Between phases						
	Percent Unbalance						
Percent Amplitude Modulation	Phase AB						
	Phase BC						
	Phase CA						
Current	Phase						
	Measured Amperes						
kW of Load							

Remarks:

FIGURE 3. Sample information for power conversion equipment - steady state test.

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POWER CONVERSION EQUIPMENT - TRANSIENT TEST - LOAD REMOVAL

SHIP NAME AND NO. _____

PCE NO. _____

	Normal	Alternate	Normal	Alternate	Normal	Alternate
400 Hz Voltage prior to removal of load						
Maximum 400 Hz Voltage during transient						
400 Hz Voltage after transient						
Maximum percent voltage rise						
Voltage recovery time						
Frequency prior to removal of load						
Maximum Transient frequency						
Frequency recover time						
400 Hz Current prior to transient — Phase Meas. Amps						
Magnitude of current during maximum transient						
Current after transient						
Maximum percent 60 Hz volt, rise						
60 Hz volt recovery time						
Load removed (name)						
Remarks:						

FIGURE 4. Sample information for power conversion equipment - transient test load removal.

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POWER CONVERSION EQUIPMENT - TRANSIENT TEST - LOAD APPLIED

SHIP NAME AND NO. _____

PCE NO. _____

	Normal	Alternate	Normal	Alternate	Normal	Alternate
400 Hz Voltage prior to application of load						
Minimum 400 Hz Voltage during transient						
400 Hz Voltage after transient						
Maximum percent voltage dip						
Voltage recovery time						
Frequency prior to addding load						
Maximum transient frequency						
Frequency recovery time						
400 Hz Current prior to transient — Phase Meas.						
400 Hz Current prior to transient — Amps.						
Magnitude of current during maximum transient						
Current after transient						
Maximum percent 60-Hz volt drop						
60 Hz volt recovery time						
Load added (name)						
Remarks						

FIGURE 5. Sample information for power conversion equipment - transient test load applied.

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POWER CONVERSION EQUIPMENT - HARMONIC VOLTAGE ANALYSIS TEST

SHIP NAME AND NO. _____

		PCE No.			PCE No.			PCE No.		
Harm. No.	Freq. (Hz)	No Load	Normal Load	Alternate Load	No Load	Normal Load	Alternate Load	No Load	Normal Load	Alternate Load
1	400									
2	800									
3	1200									
4	1600									
5	2000									
6	2400									
7	2800									
8	3200									
9	3600									
10	4000									
11	4400									
12	4800									
13	5200									
14	5600									
15	6000									
16	6400									
17	6800									
18	7200									
19	7600									
20	8000									
21	8400									
22	8800									
23	9200									
24	9600									
25	10000									
26	10400									
27	10800									
28	11200									
29	11600									
30	12000									
31	12400									
32	12800									

FIGURE 6. Sample information for power conversion equipment - harmonic voltage analysis test.

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POWER CONVERSION EQUIPMENT - HARMONIC VOLTAGE ANALYSIS TEST

SHIP NAME AND NO. _____

		PCE No.			PCE No.			PCE No.		
Harm. No.	Freq. (Hz)	No Load	Normal Load	Alternate Load	No Load	Normal Load	Alternate Load	No Load	Normal Load	Alternate Load
33	13200									
34	13600									
35	14000									
36	14400									
37	14800									
38	15200									
39	15600									
40	16000									
41	16400									
42	16800									
43	17200									
44	17600									
45	18000									
46	18400									
47	18800									
48	19200									
49	19600									
50	20000									
Phased Meas.										
Total Percent Harm.										
Remarks:										

FIGURE 6. Sample information for power conversion equipment - harmonic voltage analysis test. - Continued

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POWER CONVERSION EQUIPMENT - HARMONIC VOLTAGE TEST - THREE PHASE

SHIP NAME AND NO. _____

PCE No.

		No Load			Normal Load			Alternate Load		
Phase		AB	BC	CA	AB	BC	CA	AB	BC	CA
Harm. No.	Freq. (Hz)									
1	400									
2	800									
3	1200									
4	1600									
5	2000									
6	2400									
7	2800									
8	3200									
9	3600									
10	4000									
11	4400									
12	4800									
13	5200									
Total Percent Harmonic										

Remarks:

FIGURE 7. Sample information for power conversion equipment - harmonic voltage test - 3 phase.

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POWER CONVERSION EQUIPMENT - LINE VOLTAGE REGULATOR NO LOAD STABILITY TEST SHIP NAME AND NO. _____			
PCE No.			
PCE Swbd No.			
Percent Voltage Modulation			
Voltage of time of maximum modulation			
Remarks:			

FIGURE 8. Sample information for power conversion equipment - line voltage regulator no load stability test.

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LINE VOLTAGE REGULATOR - STEADY STATE TEST SHIP NAME AND NO. _____						
Regulator No.						
LVA Rating						
Voltage						
Phase						
	Normal	Alternate	Normal	Alternate	Normal	Alternate
Input power source						
Input voltage						
Input frequency						
Input current						
Output Voltage						
Phase						
Output Frequency						
Output Current						
Output Percent Voltage Modulation						
Remarks:						

FIGURE 9. Sample information for line voltage regulator - steady state test.

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LINE VOLTAGE REGULATOR - TRANSIENT TEST						
SHIP NAME AND NO. _____						
Regulator No.						
	RISE	DIP	RISE	DIP	RISE	DIP
Maximum voltage transient percent						
Voltage recovery time						
Maximum frequency transient percent						
Load removed and reapplied (name)						
Line current to the load						
Remarks:						

FIGURE 10. Sample information for line voltage regulator - transient test.

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LINE VOLTAGE REGULATOR - HARMONIC TEST

SHIP NAME AND NO. _____

Regulator No													
Harm. No.	Freq. (Hz)	No Load	Nor. Load	No Load	Nor. Load	No Load	Nor. Load	No Load	Nor. Load	No Load	Nor. Load	No. Load	Nor. Load
1	400												
2	800												
3	1200												
4	1600												
5	2000												
6	2400												
7	2800												
8	3200												
9	3600												
10	4000												
11	4400												
12	4800												
13	5200												
Total Percent Harmonic													
Phase Meas.													
Remarks:													

FIGURE 11. Sample information for line voltage regulator - harmonic test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-STD-2189(SH) SECTION 314-4	DESIGN METHODS FOR NAVAL SHIPBOARD SYSTEMS 400 HERTZ POWER SYSTEM TEST PROCEDURE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording

c Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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